VAIBHAV KRISHNA BANSAL

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RESEARCH PROFILE

Robotics and machine learning researcher (B.Tech., 2027) with IISc research experience in constrained trajectory planning, hands-on ROS 2 autonomy projects, vision-based closed-loop UAV control, and peer-reviewed research presentations. Interested in robot perception, motion planning, and learning for physical interaction.

EDUCATION

Manipal Institute of Technology, MAHE | B.Tech. Information Technology | 2023–2027

Relevant coursework: Machine Learning, Artificial Intelligence, Deep Learning, Neural Computation, Generative AI, Explainable AI, Design & Analysis of Algorithms, Embedded Systems, Engineering Mathematics I–IV.

RESEARCH & ROBOTICS EXPERIENCE

Research Intern | Human Interaction and Robotics (HIRO) Lab, IISc Bengaluru | May–Jul 2026

- Developed a Model-Based Diffusion planner for a non-holonomic robot, using differentiable Signal Temporal Logic (STL) robustness to optimize obstacle safety, goal reaching, and strict spatio-temporal waypoint constraints.
- Designed two navigation architectures: an offline global planner with online receding-horizon control (RHC), and a fully online RHC planner operating under partial observability with dynamic obstacles and incremental environment memory.
- Implemented time-aware local planning that brakes or loiters to satisfy waypoint arrival windows, using Model-Based Diffusion to sample and iteratively refine multiple control sequences under differentiable STL objectives.

Head of Autonomous Drone Research | AeroMIT, MIT Manipal | Jan 2024–Present

- Lead a student research subsystem working on autonomous UAV perception, navigation, and control for challenging and GPS-denied environments.
- Develop ROS 2 / MAVLink-integrated UAV workflows and Gazebo simulations; apply OpenCV and 3D scene understanding to perception-driven autonomy and system testing.

SELECTED ROBOTICS & RESEARCH PROJECTS

Autonomous Mobile Robot Exploration & Mapping | ROS 2, Gazebo, Python, URDF/SDF | Code

- Built a differential-drive autonomy pipeline with online SLAM, frontier-based exploration, and navigation through an unknown labyrinth; demonstrated a 7.5-minute autonomous mapping run.
- Diagnosed distorted SLAM maps to a wheel-separation mismatch (0.40 m configuration versus 0.36 m actual geometry); corrected odometry and recovered consistent mapping.
- Replaced a CPU-heavy RGB-D-to-laserscan pipeline with 360° GPU LiDAR, recovering control-loop frequency from 1.3 Hz to 20 Hz; tuned costmaps for narrow corridors.

Physics-Informed Sounding Rocket Flight-Phase Classification | Python, Keras, ML | Code

- Built a five-phase flight-state classifier using GPS-derived telemetry without barometric sensing; evaluated on seven rockets with Leave-One-Rocket-Out cross-validation.
- Developed PI-XGB with a physics-penalized objective, reaching 0.838 ± 0.112 mean macro-F1 versus 0.830 for baseline XGBoost.
- Designed PhaseNet with multi-scale temporal convolutions, attention fusion, and physics-informed losses; implemented streaming inference and early apogee prediction.

Autonomous Precision Landing Drone | OpenCV, ArUco, Proportional Control | Code

- Built an autonomous UAV sequence for takeoff, ArUco-marker detection, target alignment, and precision landing.
- Implemented image-based visual servoing: converted marker-position error into real-time velocity commands using proportional feedback control.

PUBLICATIONS & RESEARCH PRESENTATIONS

- V. K. Bansal et al., "Physics-Informed Gradient Boosting and Explainable AI for Real-Time Sounding Rocket Phase Classification," IC3 2026. Accepted and presented; reported as in press for IEEE Xplore.
- V. K. Bansal, "Design and Multi-Faceted Validation of UAV Arm Clamps, Landing Gear and Motor Mounts via Generative Design in Autodesk Fusion 360," SAE AeroCON 2026. Accepted and presented.
- P. Rao and V. K. Bansal, "Bandwidth-Efficient Map Sharing in Multi-Robot SLAM Using Generative Models," CASML 2025, IISc Bengaluru. Peer-reviewed; accepted for poster presentation.

TECHNICAL SKILLS

Programming: Python, C, Java, JavaScript

Machine learning: PyTorch, TensorFlow/Keras, scikit-learn, NumPy, Pandas

Robotics & computer vision: ROS 2, Gazebo, MAVLink/MAVSDK, OpenCV, URDF/SDF, 3D perception

Core methods: Robot kinematics and dynamics, path/motion planning, state estimation, Kalman filtering, Signal Temporal Logic

Tools: Linux, Git, Fusion 360

COMPETITIONS & ROBOTICS ACHIEVEMENTS

- International Space Drone Challenge 2026 — 7th internationally; team leader of a six-member team.
- e-Yantra Robotics Competition 2026, IIT Bombay — top 20 nationally; team leader; worked on PID/LQR control for physical nano-drones.
- ICUAS UAV Competition 2025 — 9th internationally with team; multi-UAV autonomy and integration.
- Quark 2025 Search & Rescue, BITS Goa — 2nd nationally with team; UAV navigation and payload delivery.
- SAE Aero Design East 2026 — team placed 4th internationally.

RELEVANT WORKSHOPS & TRAINING

Kalman Filters & Non-Linear Observers | MIT Manipal

Three-day intensive workshop led by Dr. Ravi Banawar (IIT Bombay), focused on Kalman filtering and nonlinear observer methods for dynamic-system state estimation.

Advanced Guidance, Control and State Estimation | IISc Bengaluru | Jul 2025

Two-week academic lecture series covering optimal control, guidance and navigation, dynamic programming, and state estimation, including Kalman, Unscented and Particle Filters.

Additional training: Deep Learning Specialization; Machine Learning (Andrew Ng); Python for Data Analysis — Coursera.